

ECON211: Quiz 1A

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Instructions: general

- DO NOT TURN THIS PAGE BEFORE THE EXAM STARTS!
- Write your full name above.
- You cannot use notes, books, or electronic equipment including phones, headphones, calculators, etc.
- Use scratch paper or the back of the exam pages for your computations.
- No communication or gazing into others' papers.
- You have **50 minutes**.
- There are no partial points.
- After time runs out, you must close the exam and put it in front of you **immediately**.
- If you violate any of the above rules, you fail the exam.
- Good luck!

Instructions: how to write your answers

- Your final answer must be either a number, an arithmetic or an algebraic expression, or a short sentence, depending on the question. You can leave your answer in terms of fractions, binomials, exponents, etc., that is no need to compute an exact numerical value, an arithmetic expression is sufficient.
- Your solutions must be written at the last page of the exam or on a separate signed page. I will consult your solutions in case of an ambiguity.
- Circle each of your final answers! Points are awarded for correct final answers; solutions are only consulted for clarification.
- You cannot use pencils to write down your answers.

1. I plan to distribute rewards to the 50 students in my class.

- (3 points) Suppose I have 50 *distinct* postcards. If each student is to receive at most one postcard, in how many ways can the postcards be distributed?
- (3 points) Suppose I have 50 *distinct* postcards. If there is no limit on the number of postcards a student may receive, in how many ways can the postcards be distributed?
- (3 points) Suppose I have 50 *identical* postcards. If there is no limit on the number of postcards a student may receive, in how many ways can the postcards be distributed?
- (3 points) Suppose I have one gold, one silver, and one bronze medal. If each student is to receive at most one medal, in how many ways can the medals be distributed?
- (3 points) Suppose I have three *identical* medals. If each student is to receive at most one medal, in how many ways can the medals be distributed?
- Suppose I have 10 *distinct* postcards and I distribute all of them randomly, such that each student may receive any number of postcards. Alice and Bob are students in the class. Let A be the event that Alice receives at least one postcard, and B be the event that Bob receives at least one postcard.
 - (a) (2 points) Verbally interpret the event $A^c \cap B^c$ in a single, short sentence.
 - (b) (3 points) Calculate the probability $P(A^c \cap B^c)$.
 - (c) (2 points) Are the events A and B disjoint? Briefly explain your reasoning.
 - (d) (5 points) What is the probability that both Alice and Bob receive at least one postcard?

2. There are two boxes with balls. Box 1 has 10 black and 15 white balls; Box 2 has 10 white balls only. A box is chosen at random with equal probability. You do not know which box is chosen and randomly take two balls out of the chosen box without replacement.

- (a) (2 points) Is the color of the first ball you take independent from the color of the second ball? Briefly explain.
- (b) (3 points) Given that both chosen balls are white, what is the probability that Box 1 is chosen?
- (c) (4 points) What is the probability that the chosen balls are of the same color?

Write your solutions here.